

ENAS KARAEM

Genova, Italy | +39 3497179961 | enaskrayem@gmail.com
github.com/EKaraem | linkedin/enas-karaem

Skills

Programming & Development: Python, Angular, ASP.NET, C#, TypeScript, NodeJS, SQL.

AI & Generative AI: AWS Bedrock (LLM application prototype in hackathon setting), prompt-based systems, Q&A powered by basic LLM over structured data.

DevOps & Cloud: Docker, Kubernetes, ArgoCD, GitLab CI/CD, GitHub Actions, Nginx, Linux, AWS, Proxmox, Octopus Deploy, RabbitMQ.

Collaboration & Tools: GitHub, Git, Jira, Agile workflows, team collaboration.

WORK EXPERIENCE

Software Engineer

2024 – PRESENT

Teoresi Group - Consultant at Alstom

Genova, Italy

- Developed Telediagnostica, a real-time railway analytics system using RabbitMQ microservices for high-traffic data.
- Built scalable micro-frontends using Angular Federation and TypeScript, optimizing real-time telemetry for high-frequency train monitoring.
- Implemented microservice and proxies to streamline communication between services and client infrastructure.
- Managed virtualized infrastructure via Proxmox, including VPN setup, and monitored system health, alarms, and notifications using CheckMK, improving incident response by 20%.
- Managed containerized microservices with Docker, orchestrated via Kubernetes and ArgoCD; implemented GitLab CI/CD pipelines for automated testing, integration, and security compliance; deployed releases via Octopus Deploy; and monitored system performance with Grafana to ensure operational integrity.

Technologies: ASP.NET, Angular Federation, Nginx, GitLab, Docker, Docker Compose, Kubernetes, ArgoCD, RabbitMQ, Grafana, Harbor, Linux, Proxmox, Octopus Deploy, Jira, Git.

Software Engineer

2022 – 2023

Liguria Digitale

Genova, Italy

- Optimized the LiguriaGO transportation ecosystem's backend APIs and front-end performance for high-traffic usage.
- Developed a building classification platform using React and MapLibre, managing complex geospatial datasets and improving data processing and map rendering performance by 20%.
- Leveraged the ELK Stack for system-wide log analysis, identifying performance bottlenecks and security anomalies.

Technologies: React.js, FastAPI, Python, Elasticsearch, Kibana, Jenkins, Groovy, Docker, Power BI MapLibre, Git.

Software Engineer

2020 – 2022

Geospatial Libya Consulting - ESRI Partner

Libya

- Built geospatial analytics platforms for national-scale location datasets, enabling map-based visualization, spatial filtering, and decision support for public-sector initiatives.
- Designed a real-time geospatial COVID-19 dashboard for crisis monitoring; implemented high-volume data ingestion, aggregation, and map rendering optimizations that reduced load time by 35%.

Technologies: ArcGIS, Dashboards, JavaScript, Python, QGIS, SQL.

Web Developer

2019 – 2020

Misurata University

Libya

- Architected and deployed the university's web infrastructure from scratch, including full database schema design and end-to-end migration of student and faculty records.

Technologies: PHP, JavaScript, MySQL.

EDUCATION

Master of Science (MSc) in Computer Science (Cybersecurity)

2019 – 2022

University of Genoa (Università degli Studi di Genova)

Genova, Italy

- *Thesis: Blockchain-based academic credential system (Solidity NFT)*
- Coursework: Distributed Systems, Binary Analysis, Malware Analysis, Reverse Engineering, Digital Forensics.
- *Advanced Tools: Ghidra, Autopsy, Solidity, Blockchain, DFIR, Docker, Node-RED, Python.*

Winter School: Blockchain Technology and Applications

2021

University of Camerino, Italy

Bachelor of Science (BSc) in Information Technology

2014 – 2018

Misurata University, Faculty of Information Technology

Libya